

Network Segmentation: Proxmox/OPNsense VLAN Deployment & Routing Diagnosis

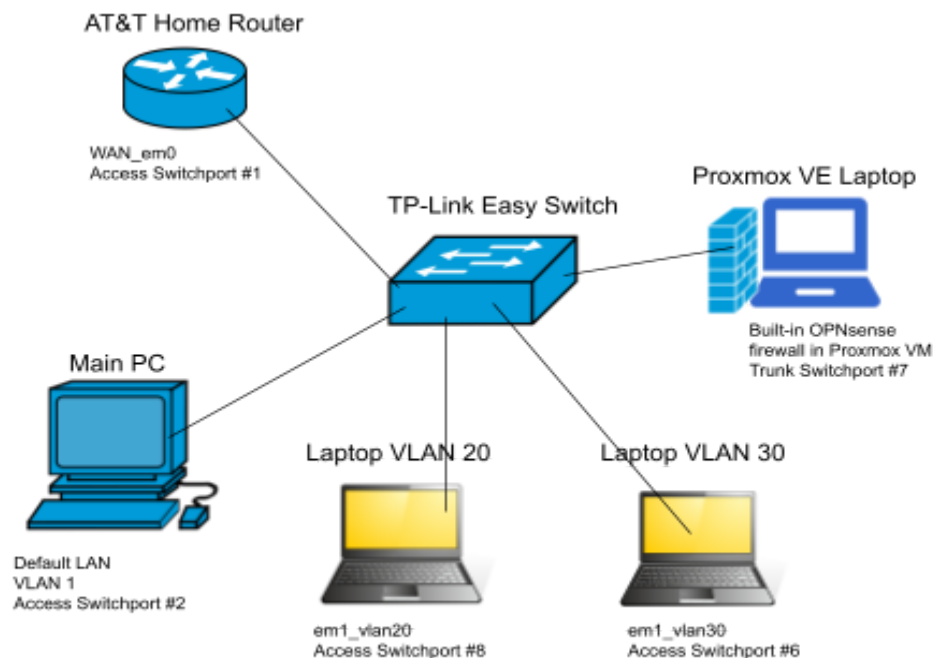
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Tools & Environment: Proxmox VE (Hypervisor) | Firewall Appliance (OPNsense VM) | TP-Link SG108E Managed Switch | Cat6 & Cat5e ethernet cables | Lenovo Thinpad Carbon X1 4th Gen | Windows 7 Toshiba Laptop (Test Device)

Overview: Deployed a Proxmox VE hypervisor on a refurbished laptop I fixed, installed OPNsense firewall and configured a multi-VLAN environment. Assigned OPNsense's virtual interface, virtual IP configuration, and reinforced VLAN IDs through TP-Link network switch. The Network Segmentation Project was designed for studying for the Network+ exam, which I am now certified for.

What This Demonstrates:

- Proxmox VE hypervisor installation and management
- OPNsense firewall VM deployment and VLAN interface assignment
- Network and VLAN connection troubleshooting
- Virtual firewall rule configuration & security practices
- Root-cause analysis through VLAN testing



Home Lab Physical Diagram Design

Full Lab Configuration- Documentation

- Extracted Proxmox VE .ISO file into a 8GB USB drive via Rufus extraction tool.
- On a separate PC, I accessed the Proxmox web GUI and typed in login credentials.
- Check interface links: > ip link

```
Welcome to the Proxmox Virtual Environment. Please use your web browser to
configure this server - connect to:

-----

proxmox login: 
Password: 
Linux proxmox: 

The programs included with the Debian GNU/Linux system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/*copyright.

Debian GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent
permitted by applicable law.
root@proxmox:~# ip link
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN mode DEFAULT group default qlen 1000
    link/loopback 
2: nic0: <BROADCAST,MULTICAST> mtu 1500 qdisc noop state DOWN mode DEFAULT group default qlen 1000
    link/ether 
    altname 
3: 
    link/ether 
4: vlp4s0: <BROADCAST,MULTICAST> mtu 1500 qdisc noop state DOWN mode DEFAULT group default qlen 1000
    link/ether 
    altname 
5: vmb0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc noqueue state UP mode DEFAULT group default qlen 1000
    link/ether 
root@proxmox:~#
```

Check Interfaces: enx.. & vmb0 interfaces = UP, nic0 & vlp4s0 = DOWN

- Configured Proxmox VE's network interfaces and web browser access using nano script

Nano Script Sample

```
auto lo
```

```
iface lo inet loopback
```

```
auto enx...
```

```
iface enx...
```

```
auto vmb0
```

```
iface vmb0 inet static
```

```
    address 192.168.X.X/24
```

```
    gateway 192.168.X.X
```

```
    bridge-ports enx...[ethernet interface name]
```

```
    bridge-stp off
```

```
    bridge-fd 0
```

```
    bridge-vlan-aware yes
```

- Type 'ifreload -a' to reload configurations

Configuring the LAN Interface

- vtnet0 parent interface
- VLAN 20 as vlan02
- VLAN 30 as vlan03.

```
Guest (vlan03) -> v4:
LAN (vtnet0) -> v4:
Lab (vlan02) -> v4:

HTTPS: sha256 [REDACTED]

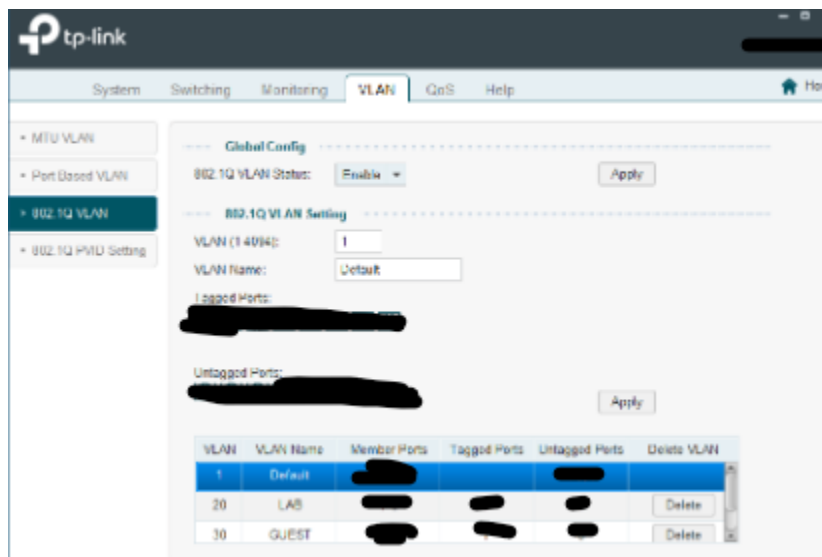
0) Logout
1) Assign interfaces
2) Set interface IP address
3) Reset the root password
4) Reset to factory defaults
5) Power off system
6) Reboot system
7) Ping host
8) Shell
9) pfTop
10) Firewall log
11) Reload all services
12) Update from console
13) Restore a backup

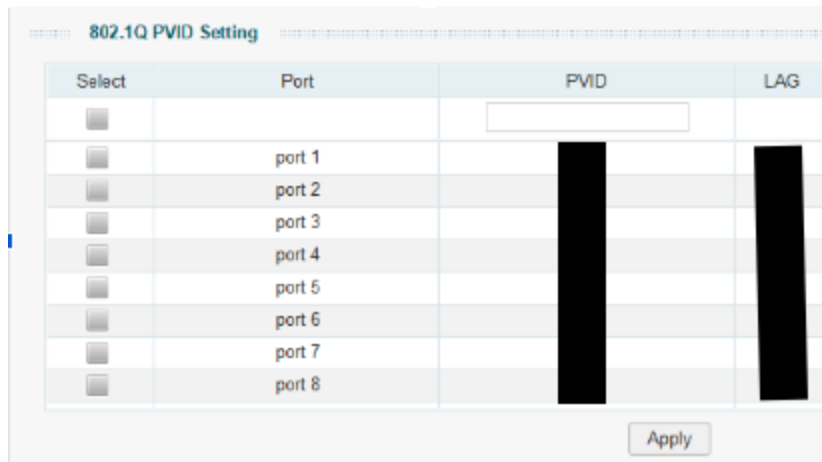
Enter an option: 
```

OPNsense Interface, IP Address, and VLANs Assigned

Network Switch Configuration

Used the Easy Smart Configuration Utility from TP-Link switch to configure and assign each VLAN with their designated ports and PVID settings.





TP-Link Easy Smart Configuration Utility- 802.1Q VLAN tags and PVID Assigned

OPNsense Firewall ACL Rules Assigned

Firewall: Rules: LAN Inspect

☐	Protocol	Source	Description	☐
Automatically generated rules 22				
☐	IPv4 *	LAN net	Default allow [redacted] to any rule	☐
☐	IPv6 *	LAN net	Default allow [redacted] IPv6 to any rule	☐
☐	IPv4 *	Lab net	Allow VLAN [redacted] out to home network	☐

Firewall: Rules: Lab Inspect

☐	Protocol	Source	Description	☐
Automatically generated rules 15				
☐	IPv4 *	Lab net	Allow [redacted] to Internet	☐
☐	IPv4 TCP/UDP	Lab net	Allow DNS from [redacted] to firewall	☐
☐	IPv4 *	Lab net	Block [redacted] to [redacted]	☐

Firewall: Rules: Guest Inspect

☐	Protocol	Source	Description	☐
Automatically generated rules 15				
☐	IPv4 *	Guest net	Allow [redacted] to Internet	☐
☐	IPv4 TCP/UDP	Guest net	Allow DNS from [redacted]	☐
☐	IPv4 *	Guest net	Block [redacted] to [redacted]	☐
☐	IPv4 *	Guest net	Block [redacted] to [redacted] net	☐

All Firewall Rules Applied on LAN (vtnet0), Lab (VLAN10), and Guest (VLAN30)

Network Switch VLAN Tagging & Segmentation- Troubleshooting & Verifying VLAN Assignment

Issue: VLAN, IP address, and subnets have been configured, but 2+ client devices cannot reach external devices on the same VLAN. VLAN-connected devices do not have an assigned VLAN tag, resulting in receiving an APIPA address instead of VLAN 20-assigned IP address. All switchports are active, network interfaces are up, and ethernet connections are properly connected.

Diagnosis: There are two likely reasons for the cause of the issue.

- Internet AT&T router is not compatible with reaching to VLAN segmented traffic. May need to add another router that has inter-vlan routing capabilities or a multilayer (Layer 3) switch that will allow the internet to reach VLAN subinterfaces devices. Perhaps it will allow me to create SVI gateways with the proper Layer3 hardware/ software.
- Firewall rules have not been specified yet, I need to assign firewall rules on OPNsense's ACL list. There is currently a default implicit deny rule and it is most likely not allowing VLAN-connected devices to receive a VLAN ID tag/ IP assignment from OPNsense or the network switch

Fix:

Previous Troubleshooting Attempts

- View Firewall Log Files
- Ping from VLAN 20 device to Internet (8.8.8.8 or access web browser) = Failed, 100% Packet
- Ping VLAN Devices: Ping from VLAN 20 device to OPNsense IP address
 - Success, 0.0% Packet Loss

Interfaces: Diagnostics: Ping

Ping											Jobs
Search											
☐	Description	Hostname	Source	Send	Received	Min	Max	Avg	loss	Error	Commands
☐		192.168.20.1	192.168.1.1	1	1	0.066	0.066	0.066	0.0%		x ■
☐		192.168.1.1	192.168.20.1	55	55	0.03	0.204	0.067	0.0%		x ■
☐		8.8.8.8	192.168.20.1	95	0				100.0%	sendto: No route to host	x ■
Showing 1 to 3 of 3 entries											

**Ping Attempt from OPNsense Firewall*

- Pinged directly from VLAN 20 device = same results.

The issue could be fixed by purchasing and deploying a new router with inter-VLAN routing capabilities or replacing the current layer 2 network switch for a multi layer (layer 3) network switch. To reduce costs and the need for additional hardware, the quickest and most efficient way

to segment the VLANs is through implementing a “router-on-a-stick” technique. This will ensure that OPNsense’s inter-VLAN routing interfaces and the current layer 2 network switch can assign VLAN tags to their proper switchport. The laptop hosting OPNsense/ Proxmox is connected to a configured tagged switchport, and the rest of the desktop devices will be connected to VLAN20 or VLAN30 for testing purposes. This will ensure proper VLAN segmentation through firewall and IEEE 802.1q configurations.

Testing Toshiba laptop was used to connect to the VLAN20 access switchport. To verify VLAN functionality, the laptop must have an IP subnet of 192.168.20.X (OPNsense configuration IP address scope of VLAN20).

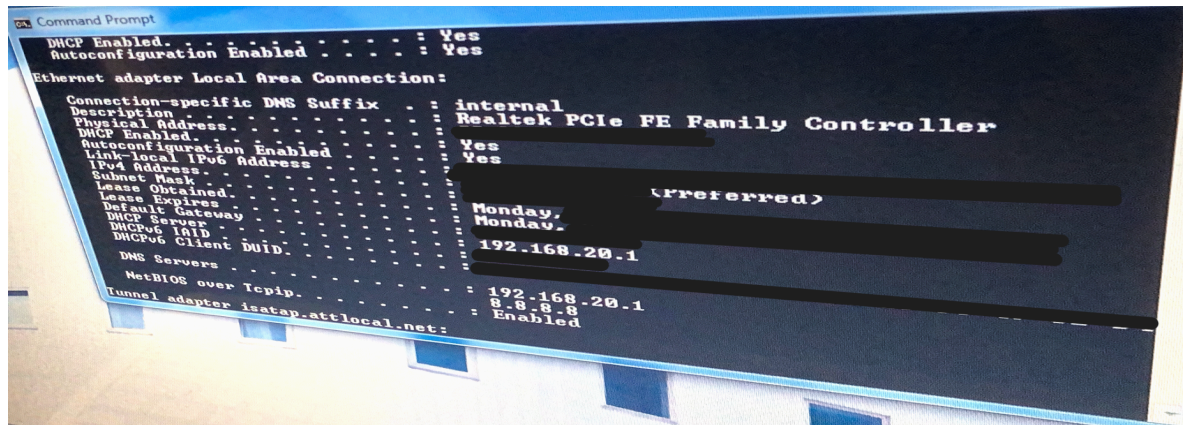
Testing VLAN Assignment by Connecting Separate Device on VLAN20 Switchport

Using my testing laptop device (Windows 7 Toshiba Laptop), I typed the “ifconfig /all” command on the Command Prompt.

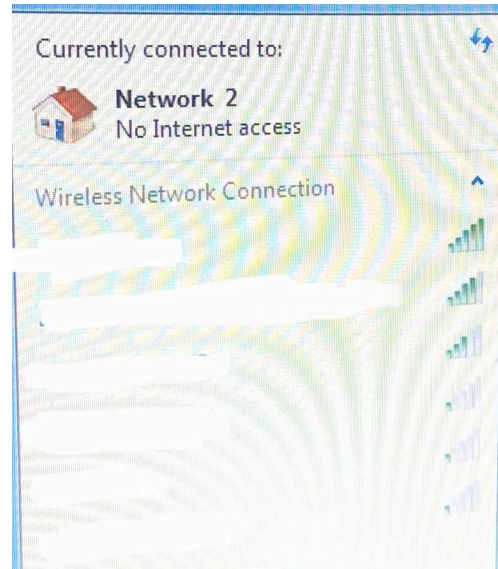
It will show the appropriate subnet configurations:

- IP address (ex: 192.168.20.50-100)
- Default Gateway (192.168.20.1) ,
- DHCP server
- DNS Server

Results



- VLAN 20/ Toshiba Laptop is connected to Network 2/ VLAN 20 along with its configured subnet IP address I have assigned on OPNsense firewall



- VLAN assignments succeed, the device was given a separate, appropriate VLAN 20 IP address when connected to access switchport
- Network 2 is VLAN 20, VLAN20 interface was assigned by OPNsense and its PVID/ VLAN ID was configured through the TP-Link network switch

Verify Full Functionality through OPNsense Firewall Logs

- Added firewall rules applied to VLAN devices to allow internet access via LAN did not make a difference. Will need to troubleshoot issues regarding VLAN devices having no internet access.
- **Current Status:** VLAN 20 is active

Interface	Time	Source	Destination	Proto	Label	
▶ LAN	→ 2026-02-			tcp	anti-lockout rule	1
▶ LAN	→ 2026-02-			tcp	anti-lockout rule	1
▶ LAN	→ 2026-02-			tcp	anti-lockout rule	1
▶ LAN	→ 2026-02-			tcp	anti-lockout rule	1
▶ Lab	← 2026-02-			udp	allow access to DHCP server	1
▶ Lab	→ 2026-02-			udp	allow access to DHCP server	1
▶ Lab	← 2026-02-			udp	allow access to DHCP server	1
▶ Lab	→ 2026-02-			udp	allow access to DHCP server	1
▶ Lab	← 2026-02-			udp	allow access to DHCP server	1
▶ Lab	→ 2026-02-			udp	allow access to DHCP server	1
▶ Lab	← 2026-02-			udp	allow access to DHCP server	1

Takeaway:

- Ensure devices connected to VLAN-assigned switchports are given an assigned IP address and subnet mask configuration
- Router-on-a-stick proved to segment VLAN traffic effectively, mainly used to reduce hardware costs
- Considering ESXi/Hyper-V for native multi-NIC VLAN support
- - Identified single-NIC hardware limitation as blocking inter-VLAN internet routing; documented resolution path (dual-NIC or Layer 3 switch upgrade) as next steps